

Session Objectives

- 1. Describe the opioid analgesic drug classes; including full and partial agonists.**
- 2. Identify the specific drugs. Correlate the use of each drug to the mechanism of action.**
- 3. Describe the specific adverse effects associated with the use of opioids.**
- 4. Identify how differences in pharmacokinetics can enhance adverse effects.**
- 5. Recognize the signs and symptoms of an opioid overdose. Differentiate an opioid overdose from withdrawal.**
- 6. Describe the critical use of opioid antagonists in overdose situations.**

Analgesia

- **Strong agonists: morphine, methadone, heroin, meperidine, fentanyl**
- **Mixed agonist-antagonist: buprenorphine**
- **Weak (or partial) agonist: codeine, hydrocodone (often combined with COX inhibitor).**

Cardio-Respiratory

- **Vasodilation= hypotension**
- **Reduction of myocardial oxygen demand, cardiac work.**
- **Decrease PVR, preload, afterload**
- **Cause of death in overdose =Respiratory depression**
- **Opioids decrease the sensitivity of the medulla to increased carbon dioxide.**
- **Increase intracranial pressure (worsen head injury outcome).**

Neonatal Abstinence Syndrome

- The fetus may become physically dependent in utero and manifest withdrawal symptoms early postpartum
- Withdrawal symptoms in the infant: irritability, shrill crying, diarrhea, or seizures.
- More severe withdrawal: camphorated tincture of opium (previous). Current: Morphine, methadone
- Phenobarbital and clonidine (adjunctive use)

Opioid Withdrawal

- Occurs 6-12 h after last dose, continues for several days, and ends within 72 h.
- Symptoms: Rhinorrhea, sweating, lacrimation, chills, gooseflesh, yawning, anxiety, aggression, mydriasis
- Drugs with a long $t_{1/2}$, produce a prolonged withdrawal (Methadone)
- Opioids bind in the nucleus accumbens, release dopamine leading to addiction “reward center”

Morphine

- “Standard” opioid agonist
- Reacts with all opioid receptors
- Uses: moderate-severe pain, reduces dose of anesthetics, and anxiolytic effect.
- MI: produces vasodilation
- Hypersensitivity in respiratory disease
- **Respiratory depression (fatal)**
- Not for head trauma patients
- Intense sedation and respiratory depression when combined with other **sedatives, alcohol, or barbiturates.**

Codeine

- Mild-moderate pain; metabolized to morphine. Considered to have “partial” activity.
- Combined with non-opioids to relieve pain.
- Antitussive
- Codeine undergoes **CYP2D6** metabolism to morphine. Genetic polymorphisms may increase effects.
- “Ultra-fast metabolizer” is a concern.

Meperidine

- **Rapid onset-short duration of action.**
- **Normeperidine (metabolite) can induce CNS stimulation.**
- **Serotonin syndrome with SSRIs, linezolid, tricyclic antidepressants, “triptan” drugs for migraines.**
- **Not with monoamine oxidase inhibitors (MAOIs)**
- **Anti-muscarinic effects (mydriasis, tachycardia)**
- **Some local anesthetic effects**

Levorphanol

- **L-isomer of morphine**
- **Moderate to severe pain**
- **D-isomer (Dextromethorphan; DM). Not addictive
cough suppression at normal doses.**
 - **NMDA receptor antagonist**
 - **SERT and NET reuptake inhibition**

Heroin: Schedule I

- **Injected, snorted, adulterated.**
- **Chronic injection leads to collapse of vessels**
- **Highly lipophilic**
- **Metabolized to morphine ~diacetylmorphine**
- **Pregnancy: newborn addicted to heroin, withdrawal.**
- **Increased incidence of HIV and hepatitis (needle sharing)**
- **Oxycontin[®]- gave a “heroin like” effect when crushed and snorted. New formulations provide special coatings to prevent crushing.**

Tramadol

- μ receptor agonist and reuptake inhibitor of NE and 5HT
- Can precipitate withdrawal.
- CYP2D6 polymorphism “ultra-fast metabolism”
- Used for chronic pain (neuropathic)
- Combined with acetaminophen/NSAIDs
- Tapentadol (μ receptor agonist and reuptake inhibitor of NE, minimal effect on 5-HT).
- Both have serotonin syndrome warning.

Methylnaltrexone

- Treats opioid-induced constipation.
- Does not cross blood brain barrier (BBB). Quaternary ammonium cation
- Alvimopan: treats postoperative ileus. Antagonist of mu receptor in GI tract. Contraindicated in patients who had opioids within 7 days; do not give alvimopan for more than 7 days. MI risk increased.

Mixed Agonist-Antagonist

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- Pentazocine, Butorphanol, Nalbuphine
- Retains analgesia with reduced addiction potential ($\kappa > \mu$ receptor).
- Less physical dependence and respiratory depression.
- Psychomimetic effects (κ receptor). Pentazocine
- **Precipitate withdrawal in morphine addicted patient.**

Buprenorphine

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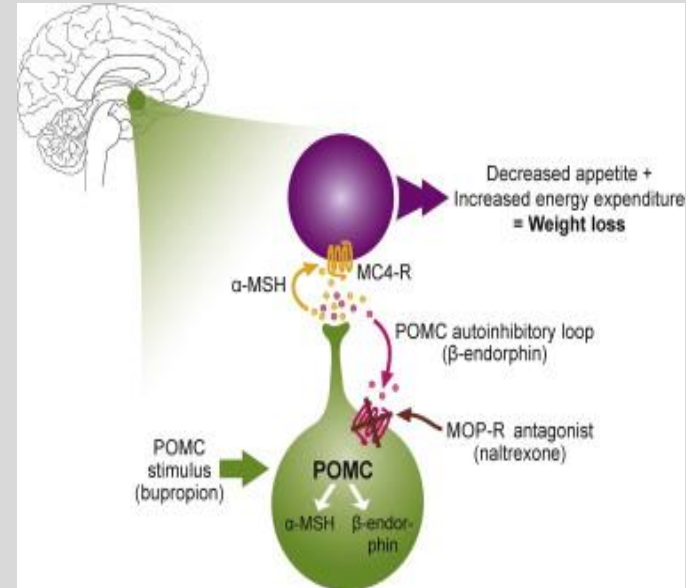
- Partial μ agonist. At higher doses is an antagonist because of K receptor blockade
- Slow dissociation from μ receptor.
- CYP 3A4 substrate
- Sublingual, transdermal, parenteral
- Adverse effects are similar to morphine and pentazocine
- Treats opioid dependence
- Precipitate withdrawal in opioid users
- Buprenorphine and naloxone (Suboxone[®])

Opioid Antagonists

- **Naloxone: IV; fast onset for opioid overdose (1-3 min). Rapid offset. Repeated administration until opioid is cleared.**
- **Approved in neonates with respiratory depression born to heroin mothers**
- **Naltrexone: oral; long-duration. Maintenance therapy. Reduces alcohol cravings. Combined with bupropion for weight loss potential.**

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Opioid Abuse

- Heroin: Rapid tolerance and dependence
- MOA: μ and κ receptor agonism
- Overdose: IV naloxone and ventilation
- Symptoms: “everything is down”, stupor, bradycardia, decreased respirations, miosis.
- Withdrawal: Methadone or buprenorphine
- Symptoms: lacrimation, rhinorrhea, yawning, sweating, gooseflesh, tremors.

Suzetrigine

- **Non-opioid**
- **MOA: Sodium channel blocker, works in periphery**
- **Use: moderate-to-severe acute pain, not more than 14 days.**
- **Adverse effects: itch, muscle spasms (increased creatine kinase), N/V.**

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Suzetrigine is a potent and selective inhibitor of NaV1.8

